Dazhong Shen

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RESEARCH INTERESTS

My general research interest includes data mining and machine learning, focusing on generative models, variational inference, and their data intensive applications.

EDUCATION

Ph.D. Sep. 2017 $\sim$ June 2023
Computer Science, University of Science and Technology of China (USTC)
Advisor: Prof. Hui Xiong, AAAS/IEEE Fellow, and Prof. Enhong Chen, IEEE Fellow
Major: Computer Science in Application

B.S. Sep. 2013 $\sim$ Jul. 2017
School of Mathematical Sciences, University of Science and Technology of China (USTC)
Major: Computational Mathematics

PROFESSIONAL EXPERIENCE

Researcher, Shanghai Artificial Intelligence Laboratory
July 2023 $\sim$ Now
▷ Focused on the algorithm development for Text-to-Image Diffusion Models. And, collaborated closely with SenseTime to support the AIGC service, i.e., **Miaohua** (miaohua.sensetime.com).

- Introduced a semantic-aware classifier-free guidance strategy for the inference process to improve image quality, which supports **the 0.4.0 version of Miaohua**. (CVPR-2024)
- Pioneered the distillation technique for high-resolution image generative models (1024$\times$1024), achieving **10x inference acceleration (1.2s for one sample)**, which supports **the Turbo version of Miaohua**.
- Proposed a GAN-based optimization strategy and trained a **diffusion model with 1.7B parameter sizes** by utilizing **600+ GPUs**, i.e., **the 0.5.0 version of Miaohua**, where its image quality surpasses that of Stable Diffusion 3 based on the internal evaluation.

Research Intern, Talent Intelligence Center, Baidu Inc.
Mar. 2017 $\sim$ Sep. 2017 and Aug. 2018 $\sim$ June 2022, Mentor: Dr. Hengshu Zhu
▷ Focused on research in text representation and generation, which also supports the practical applications in the field of intelligent talent management.

- Developed a novel neural topic model with Graph Neural Networks to enhance the interpretability of text representation. (NeurIPS-2021)
- Proposed a simple but effective strategy to avoid the posterior collapse in Variational AutoEncoder (VAE) for text generation. (IJCAI-2021)
- Developed two topic models (traditional and neural) to jointly represent multiple textual collections in the scenario of talent recruitment, which has been **launched in Baidu campus recruitment system to enhance talent searching, person-job fitting, and interview question recommendation**. (IJCAI-2018 and ACM TOIS)
- Developed a Bayesian generative model to uncover the correlation between employee personality and their behavior, empowering talent management. (TMIS-2023)

Data Contest, Aftershock Detection Contest, Directed by China Earthquake Administration
May 2017 $\sim$ Oct. 2017
▷ Developed an ensemble learning-based framework for identifying seismic P-phase arrivals automatically of the aftershocks. And, we won **the championship in the First Season and the runner-up in the Finals from 1,143 teams participated from around the world**.

Research Assistant, The Hong Kong University of Science and Technology (Guangzhou)
Oct. 2022 $\sim$ Mar. 2023, Advisor: Prof. Hui Xiong
▷ Developed novel generative models for graphs with diffusion models.

AWARDS

- Outstanding Ph.D. Graduates. 2023
- China National Scholarship. (for Ph.D. students) 2022
- Huawei Scholarship (USTC). (for postgraduate and Ph.D. students) 2018, 2021
- Global Digital Technology Scholarship. 2018
- Huawei Scholarship (USTC). (for postgraduate) 2018
- IJCAI-2018 Student Travel Grants. 2018
- Aftershock Detection Contest (China Earthquake Administration), Runner-Up. 2017
- Academic Scholarship (USTC), Grad 1. six times from 2017 to 2022
- Outstanding Graduates (USTC) 2017
- Zhang Zongzhi Science and Technology Scholarship (USTC). 2016
- Outstanding Student Scholarship (USTC), Silver Medal. 2015
- Outstanding Student Scholarship (USTC), Bronze Medal. 2014

PUBLICATIONS (12 CCF A, 1 CCF A Chinese Journal)

1. **Dazhong Shen**, Guanglu Song, Zeyue Xue, Fu-Yun Wang, Yu Liu. Rethinking the Spatial Inconsistency in Classifier-Free Diffusion Guidance, In Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition **(CVPR-2024)**, 2024. **(CCF A)**
2. **Dazhong Shen**, Hengshu Zhu, Keli Xiao, Xi Zhang, Hui Xiong. Exploiting Connections among Personality, Job Position, and Work Behavior: Evidence from Joint Bayesian Learning. ACM Transactions on Management Information Systems **(TMIS)**, 2023.
3. **Dazhong Shen**, Chuan Qin, Hengshu Zhu, Tong Xu, Enhong Chen, Hui Xiong. Joint Representation Learning with Relation-enhanced Topic Models for Intelligent Job Interview Assessment, ACM Transactions on Information Systems **(ACM TOIS)**, 2021. **(CCF A)**
4. **Dazhong Shen**, Chuan Qin, Chao Wang, Zheng Dong, Hengshu Zhu, Hui Xiong. Topic Modeling Revisited: A Document Graph-based Neural Network Perspective. The 35th Conference on Neural Information Processing Systems **(NeurIPS-2021)**, 2021. **(CCF A)**
5. **Dazhong Shen**, Chuan Qin, Chao Wang, Hengshu Zhu, Enhong Chen, Hui Xiong. Regularizing Variational Autoencoder with Diversity and Uncertainty Awareness. The 30th International Joint Conference on Artificial Intelligence **(IJCAI-2021)**, 2021. **(CCF A)**
6. 申大忠, 张琦, 徐童, 祝恒书, 赵雯佳, 殷子凯, 周培伦, 房立华, 陈恩红, 熊辉. EL-Picker：基于集成学习的余震P波初动实时拾取方法, 《中国科学: 信息科学》, 2020. **(CCF A Chinese)**

7. **Dazhong Shen**, Hengshu Zhu, Chen Zhu, Tong Xu, Chao Ma, Hui Xiong. A Joint Learning Approach to Intelligent Job Interview Assessment, The 27th International Joint Conference on Artificial Intelligence **(IJCAI-2018)**, 2018. **(CCF A)**
8. Leilei Ding, **Dazhong Shen**, Chao Wang, Tianfu Wang, Le Zhang, Hui Xiong, and Yanyong Zhang. DGR: A General Graph Desmoothing Framework for Recommendation via Global and Local Perspectives, The 33rd International Joint Conference on Artificial Intelligence **(IJCAI-2024)**, 2024. **(CCF A)**
9. Wei Wu, Chao Wang, **Dazhong Shen**, Chuan Qin, Liyi Chen, and Hui Xiong. AFDGCF: Adaptive Feature De-correlation Graph Collaborative Filtering for Recommendations, The 47th International ACM SIGIR Conference on Research and Development in Information Retrieval **(SIGIR-2024)**, 2024. **(CCF A)**
10. Xiaoshan Yu, Chuan Qin, **Dazhong Shen**, Haiping Ma, Le Zhang, Xingyi Zhang, Hengshu Zhu, Hui Xiong. RDGT: Enhancing Group Cognitive Diagnosis with Relation-Guided Dual-Side Graph Transformer, IEEE Transactions on Knowledge and Data Engineering **(IEEE TKDE)**, 2024. **(CCF A)**
11. Hanzhe Li, Jingjing Gu, Xinjiang Lu, **Dazhong Shen**, Yuting Liu, Yanan Deng, Guoliang Shi, Hui Xiong. Beyond Relevance: Factor-level Causal Explanation for User Travel Decisions with Counterfactual Data Augmentation, ACM Transactions on Information Systems**(ACM TOIS)**, 2024. **(CCF A)**
12. Feihu Jiang, Chuan Qin, Jingshuai Zhang, Kaichun Yao, Xi Chen, **Dazhong Shen**, Chen Zhu, Hengshu Zhu, Hui Xiong. Towards Efficient Resume Understanding: A Multi-Granularity Multi-Modal Pre-Training Approach, The IEEE International Conference on Multimedia and Expo **(ICME-2024)**, 2024. **(CCF B)**
13. Yunfei Zhang, Chuan Qin, **Dazhong Shen**, Haiping Ma, Le Zhang, Xingyi Zhang, Hengshu Zhu. ReliCD: A Reliable Cognitive Diagnosis Framework with Confidence Awareness, IEEE International Conference on Data Mining **(ICDM-2023)**, 2023. **(CCF B)**
14. Rui Zha, Chuan Qin, Le Zhang, **Dazhong Shen**, Tong Xu, Hengshu Zhu, Enhong Chen, Career Mobility Analysis with Uncertainty-aware Graph Autoencoders: A Job Title Transition Perspective, In IEEE Transactions on Computational Social Systems **(IEEE TCSS)**, 2023. **(CCF C)**
15. Qingxin Meng, Keli Xiao, **Dazhong Shen**, Hengshu Zhu, Hui Xiong, Fine-Grained Job Salary Benchmarking with a Nonparametric Dirichlet-process-based Latent Factor Model. INFORMS Journal on Computing **(JOC)**, 2022. **(UTD-24)**
16. Chuan Qin, Kaichun Yao, Hengshu Zhu, Tong Xu, **Dazhong Shen**, Enhong Chen, Hui Xiong. Towards Automatic Job Description Generation with Capability-Aware Neural Networks, IEEE Transactions on Knowledge and Data Engineering **(IEEE TKDE)**, 2022. **(CCF A)**
17. Zhi Zheng, Chao Wang, Tong Xu, **Dazhong Shen**, Penggang Qin, Xiangyu Zhao, Baoxing Huai, Xian Wu, Enhong Chen. Interaction-aware Drug Package Recommendation via Policy Gradient. ACM Transactions on Information Systems **(ACM TOIS)**, 2022. **(CCF A)**
18. Zhi Zheng, Chao Wang, Tong Xu, **Dazhong Shen**, Penggang Qin, Baoxing Huai, Tongzhu Liu, Enhong Chen. Drug Package Recommendation via Interaction-aware Graph Induction. The 30th International World Wide Web Conference **(WWW-2021)**, 2021. **(CCF A)**

Preprint:

19. Dongzhi Jiang, Guanglu Song, Xiaoshi Wu, Renrui Zhang, **Dazhong Shen**, Zhuofan Zong, Yu Liu, and Hongsheng Li. CoMat: Aligning Text-to-Image Diffusion Model with Image-to-Text Concept Matching. arXiv preprint arXiv:2404.03653 (2024).
20. Zhuofan Zong, Bingqi Ma, **Dazhong Shen**, Guanglu Song, Hao Shao, Dongzhi Jiang, Hongsheng Li, and Yu Liu. MoVA: Adapting Mixture of Vision Experts to Multimodal Context. arXiv preprint arXiv:2404.13046 (2024).
21. Chao Wang, Hengshu Zhu, **Dazhong Shen**, and Hui Xiong. Preference or Intent? Double Disentangled Collaborative Filtering. arXiv preprint arXiv:2305.11084 (2023).
22. Chuan Qin, Le Zhang, Rui Zha, **Dazhong Shen**, Qi Zhang, Ying Sun, Chen Zhu, Hengshu Zhu, and Hui Xiong. A comprehensive survey of artificial intelligence techniques for talent analytics. arXiv preprint arXiv:2307.03195 (2023).

PATENTS

1. 申大忠，祝恒书，马超，朱琛，熊辉. 基于人工智能的辅助面试方法、装置及存储介质：中国，CN110472647[B]. 2022-06-24.
2. 申大忠，秦川，王超，董政，祝恒书，熊辉. 文本主题的确定方法、装置及电子设备: 中国，CN114492393[A]. 2022-01-17.
3. 秦川，姚开春，祝恒书，马超，申大忠. 一种招聘岗位描述文本的生成方法、装置、设备及介质: 中国，CN113627135[A]. 2020-05-08.
4. 苏昱涵，秦川，申大忠，赵洪科，宋欣，祝恒书. 用于处理技能信息的方法、模型训练方法及装置: 中国，CN114330333[A]. 2021-12-29.
5. 查睿，秦川，张乐，申大忠，徐童，祝恒书. 职业流动关系的预测方法、系统、设备及存储介质: 中国，CN114331380[A]. 2021-12-32.
6. 徐童, 郑值, 王超, 申大忠, 秦鹏刚, 童贵显, 陈恩红. 药品包推荐方法、装置、计算机系统及可读存储介质：中国，CN112802573[A]，2021-01-26.
7. 李满伟，秦川，申大忠，马海平，祝恒书，张敬帅，姚开春. 联合能力模型的训练方法、能力需求匹配方法和装置：中国，CN115330142[A]. 2022-07-25.
8. Chuan Qin, Kaichun Yao, Hengshu Zhu, Chao Ma, **Dazhong Shen**. Method, apparatus, device and medium for generating recruitment position description text: US, US20210216726[A]. 2021-05-26.

ACADEMIC SERVICES

Program Committees Member:

- International Conference on Machine Learning (ICML-2022, ICML-2023, ICML-2024)
- ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD-2022)
- Conference on Neural Information Processing Systems (NeurIPS-2022, NeurIPS-2023)
- AAAI Conference on Artificial Intelligence (AAAI-2023, AAAI-2024)
- The ACM Web Conference (WWW-2024)
- International Conference on Learning Representations (ICLR-2024)

Sub-reviewer for Journals and Conferences:

- IEEE TKDE, ACM TOIS, IEEE TSMC, etc.
- ICML, NeurIPS, KDD, WWW, SIGIR, AAAI, IJCAI, CVPR, ICDM, CIKM, WSDM, etc.